

MOBILE APPLICATION DEVELOPMENT

Course Title:	Mobile Application Development	
Course Code:	23MCAG305	
Credits Units: 03	Contact Hours: 45	L-T-P-E: 3-0-0-3
IA: UE Weightage – 50 : 50	Pass Marks (IA & UE) – 40 UE (Min: 18)	Aggregate Pass Marks: 50%
UE Question Paper Marks: 50		

Course Objectives:

COB1	To explain the basics of mobile application development.
COB2	To develop Android applications with User interface and Web Services concepts.

Course Outcomes:

At the end of the course, students will be able to

Course Outcome	Description	Bloom's Taxonomy Level
CO 1	Interpret the basic concepts of Mobile application.	2 (Understanding)
CO 2	Experiment the concepts of Activities, Fragments, and Intents through app development.	3 (Applying)
CO 3	Develop a simplified user interface design for a sample application.	3 (Applying)
CO 4	Examine the concepts of content providers with a real time application.	4 (Analyzing)
CO 5	Build a mobile application using the concepts of web services using a sample application.	5 (Evaluating)

Syllabus:

Module (Hrs)	Details	COs	Tools/Software/ Database Used
I (9 Hrs)	Mobile Application: Mobile Application Development - Mobile Applications and Device Platforms - Alternatives for Building Mobile Apps - Native vs. Hybrid Applications - Mobile Application Development Lifecycle- Mobile Application Front End-Mobile Application Back End-Key Mobile Application Services- Android Programming: Android versions –Features-Tools-Launching the Android Application-Exploring the IDE-Debugging and Publishing the Application.	CO1	Android Studio
II (9 Hrs)	Activities, Fragments, and Intents: Understanding Activities-Linking Activities Using Intents-Fragments-Displaying Notifications-Understanding the Components of a Screen-Adapting to Display Orientation-Managing Changes to Screen Orientation- Utilizing the Action Bar-Creating the User Interface Programmatically Listening for UI Notifications	CO2	Android Studio
III (9 Hrs)	User Interfaces: Using Basic Views-Using Picker Views -Using List Views to Display Long Lists-Understanding Specialized Fragments - Using Image Views to Display Pictures -Using Menus with Views- Using Web View- Saving and Loading User Preferences-Persisting Data to Files-Creating and Using Databases.	CO3	Android Studio
IV (9 Hrs)	Content Providers: Sharing Data in Android-Creating the Content Providers -Using the Content Provider-SMS Messaging -Sending Email-Displaying Maps-Getting Location Data- Monitoring a Location.	CO4	Android Studio
V (9 Hrs)	Networking and Services: Consuming Web Services Using HTTP-Consuming JSON Services- Creating Services - Binding Activities to Services -Understanding Threading.	CO5	Android Studio

TEXT BOOKS:

- 1) Jerome DiMarzio(2017), “Beginning Android Programming with Android Studio”, Fourth Edition,Wiley(Modules : 1,2,3,4,5).
- 2) Neil Smyth(2017) , “Android Studio 3.0 Development Essentials: Android”, Eighth Edition,Amazon Digital Services(Modules : 3,4).

REFERENCE BOOKS:

- 1) Dawn Griffiths, David Griffiths(2017), “Head First Android Development: A Brain-Friendly Guide”, Second Edition, O'Reilly.
- 2) Pradeep Kothari(2014), “Android Application Development ”, Black Book Kindle Edition, Dreamtech Press.